



Universitat
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UPF-TALN at the CLEF 2026 SimpleText Track: Ensemble Approach for Scientific Text Simplification and Hallucination Classification

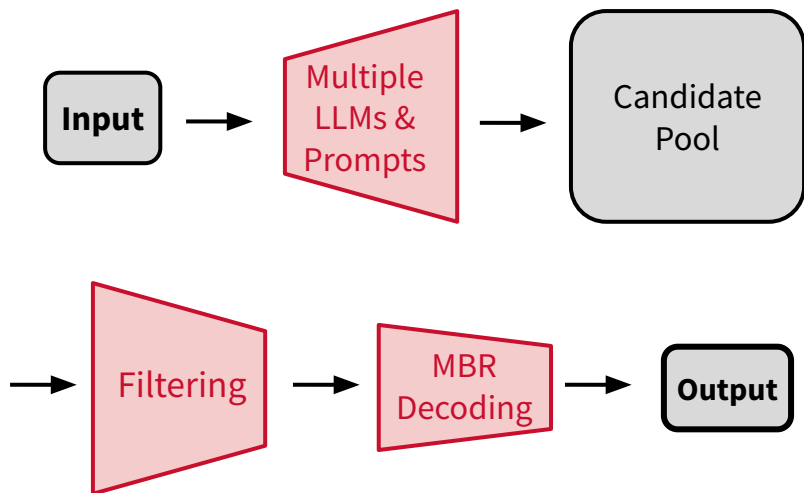
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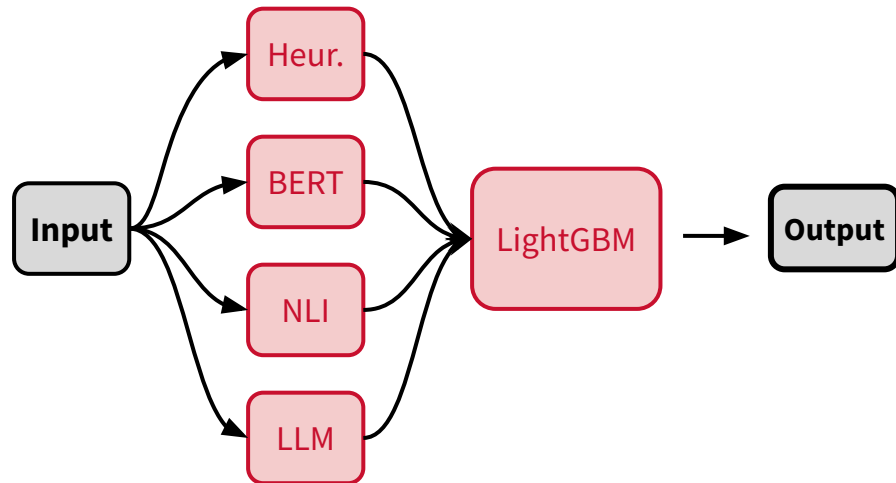
❖ Tasks we participated in

- **Task 1** - Text Simplification: Simplify Scientific Text
 - **Task 1.2:** Document-level Scientific Text Simplification
- **Task 2** - Controlled Creativity: Identify and Avoid Hallucination
 - **Task 2.2:** Detect and Classify Information Distortion Errors in Simplified Sentences

❖ Task 1.2

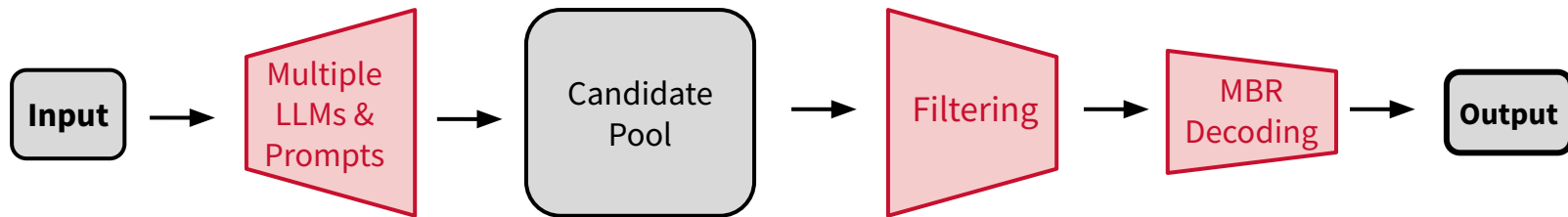


❖ Task 2.2



➤ Common idea: **Can ensembles provide more reliable outputs?**

❖ Generate-and-select Approach



- 3 prompts
- 4 LLMs
 - Gemma3-4B
 - Qwen3-4B
 - ★ GPT5-mini
 - ★ Gemini2.5-FL
- 3 generations

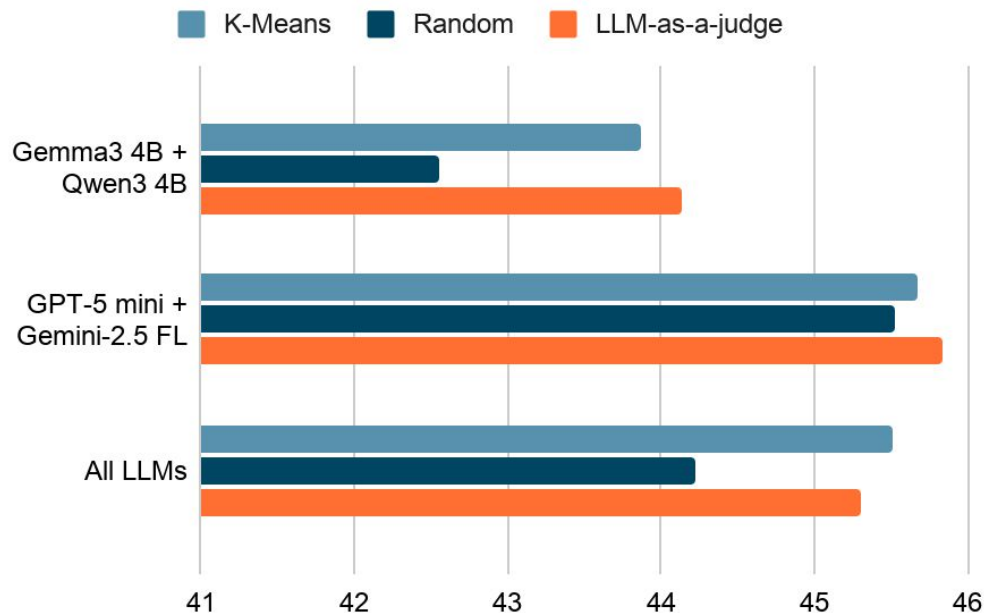
- 18 cand.
- ★ 18 cand.
- ★ 36 cand.

- Main
 - K-Means
 - Diversity (BLEU)
- ❖ Random
- ❖ LLM-as-a-judge

- Selecting the candidate most similar to the rest in SARI

Task 1.2: Results

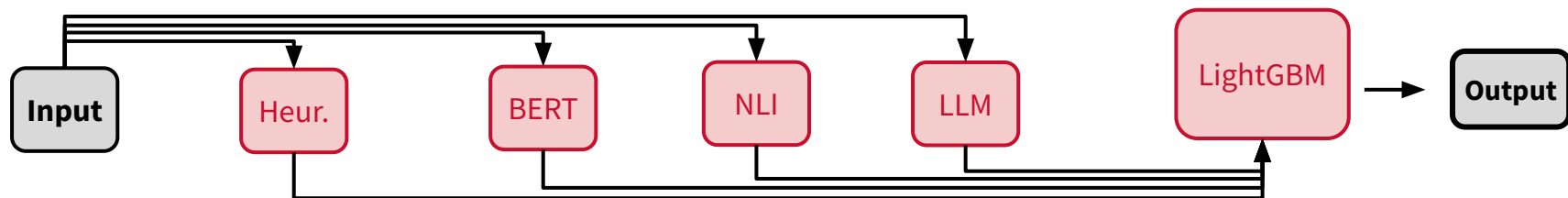
SARI Scores on CodaBench



- Rank: 5th / 18 Teams (on the final subset)
- Ensembling helps, but modestly, despite its computational cost
- **Candidate quality matters more**
 - Small LLMs show lower SARI
 - Large room for improvement in prompts?

Task 2.2: Approach

❖ Ensembling various features



- | | | | | |
|---|--|---|---|--|
| <ul style="list-style-type: none">• 39 feats.• # words, sents, ...• “would you”, “please” | <ul style="list-style-type: none">• 13 feats.• fine-tuned Clinical ModernBERT• logits, probs, category | <ul style="list-style-type: none">• 3 feats.• PubMed BERT-mnli | <ul style="list-style-type: none">• 13 feats.• Gemma3 14B-based classification• logits, probs, category | <ul style="list-style-type: none">• 68 feats. in total• 6-class classification• training with 2 different subsets
(a): 7850 instances
(b): (a) + 2000 None |
|---|--|---|---|--|

Task 2.2: Results

Training Subset	(a)	(b) (more None)
Val F1 (6-class Macro)	.8659	.8552
Test F1 (on non-None)	.7069	.2813
% Predicted None on Test	.698	.971

⌘ Full model performance

- Val performance was reasonably strong, with large gaps between Test performance
 - The text distribution appears different between Val/Test
- Key failure mode on Test was a strong bias towards None, especially for subset (b)

Task 2.2: Ablation

Training Subset	(a)	(b) (more None)
Val F1 (6-class Macro)	.8659 .8459	.8552 .8513
Test F1 (on non-None)	.7069 .7801	.2813 .7682
% Predicted None on Test	.698 .801	.971 .846

✂ Full model performance

Full w/o LLM feats. performance

- Removing LLM features improved test performance
 - Rank: 6th / 11 Teams
- Inconsistency in feature usefulness between Val/Test
 - Possible validation-test distribution mismatch

❖ Discussion

- **Ensemble $\neq$ Robustness**

- Task 1.2: Using more candidates was not very beneficial
- Task 2.2: Combining more feature sources did not necessarily improve performance

- **What we learned**

- Quality of components matters
- Distribution shift matters

Filtering	LLMs				# Cands. / Doc.	MBR Score	Final SARI
	Gemma-3 4B	Qwen-3 4B	GPT-5 mini	Gemini-2.5 Flash-lite			
K-means	✓	✓			18	48.66	43.87
			✓	✓	18	49.82	45.67
	✓	✓	✓	✓	36	47.74	45.51
Random	✓	✓			1	-	42.56
			✓	✓	1	-	45.52
	✓	✓	✓	✓	1	-	44.23
LLM-as-a-judge Ranking	✓	✓			18	49.70	44.14
			✓	✓	18	51.00	45.84
	✓	✓	✓	✓	36	48.59	45.30

Category	% in Train	Features	Meta-classifier		
		Fine-tuning BERT	Training (a)	Evaluating (b)	
None	92.33	3000	3000	5000	500
LEAKED INSTRUCTIONS	2.35	1000	1000	1000	200
UNGROUNDED INJECTION	1.64	2000	2000	2000	200
GENERATION FAILURE	3.23	1500	1500	1500	300
REPETITIVE CONTENT	0.42	300	300	300	50
GROUNDED OVERGENERATION	0.03	50	50	50	3

Train Subset	Feats.	F1 score on Val							Val Acc	Test Prediction Distribution							Test Acc
		None	LEAK	UNG	GEN	REP	OVER	(None)		LEAK	UNG	GEN	REP	OVER			
(a)	Full	.929	.870	.810	.942	.577	.500	.8659	(.698)	.180	.056	.037	.020	.009	.7069		
	- Heur.	.852	.870	.673	.946	.484	.000	.8274	(.145)	.246	.020	.040	.548	.001	.6306		
	- BERT	.928	.855	.753	.947	.590	.182	.8367	(.390)	.462	.003	.046	.021	.079	.5998		
	- NLI	.926	.873	.783	.947	.541	.333	.8566	(.656)	.212	.055	.041	.020	.017	.7063		
	- LLM	.922	.878	.776	.942	.518	.000	.8459	(.801)	.061	.090	.032	.010	.007	.7801		
(b)	Full	.925	.863	.780	.942	.569	.333	.8552	(.971)	.010	.002	.010	.007	.000	.2813		
	- Heur.	.828	.853	.675	.940	.458	.000	.8367	(.982)	.011	.002	.004	.002	.000	.1497		
	- BERT	.925	.856	.758	.940	.629	.222	.8473	(.971)	.009	.002	.010	.008	.000	.2618		
	- NLI	.922	.865	.772	.944	.605	.250	.8579	(.970)	.011	.002	.009	.008	.000	.2653		
	- LLM	.923	.871	.747	.942	.528	.000	.8313	(.846)	.049	.060	.022	.011	.012	.7682		